

4. With increase in the distance from the nucleus, velocity of electron decreases and hence, kinetic energy decreases. Potential energy in the second orbit is more than that of the first orbit since loss of energy of the electron approached from the infinite distance is less in the second orbit.

Hence, the correct option is (c).

5. According to Bohr's theory, $mvr = \frac{nh}{2\pi}$ where n is an integer. The value of $\frac{3h}{4\pi}$ is not an integral multiple of $\frac{h}{2\pi}$.

Hence, the correct option is (d).

6. The number of neutrons = 26

The number of protons = 26

The number of electrons = 26

Electronic configuration of the element is 2, 8, 14, 2.

The number of electrons present in 'M' shell of M^{+2} ion $\rightarrow$ 2, 8, 14, M^{+2} ion $\rightarrow$ 8

Hence, the correct option is (b).

7. (i) $\rightarrow$ (C); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (B)

Hence, the correct option is (a).

8. Electronic configuration of B is 2, 8, 18, 10, 2.

The number of core electrons = 38

Hence, the correct option is (c).

9. The number of protons = $\frac{14,696}{1,837} = 8$

The number of neutrons = 10

The mass number of the element = $8 + 10 = 18$.

Hence, the correct option is (c).

10. Energy of electron in Bohr's orbit of 'H' atom = $\frac{-13.6}{n^2} \rightarrow Z^2$

Energy of electron in Bohr's orbit of He^+ ion =

$$\frac{-13.6}{n^2} \rightarrow 2^2$$

$$\text{2nd orbit of H atom} \rightarrow \frac{-13.6}{2^2} = -3.4 \text{ eV}$$

$$\text{4th orbit of } He^+ \text{ ion} \rightarrow \frac{-13.6}{4^2} \rightarrow 22 = -3.4 \text{ eV}$$

Hence, the correct option is (c).

11. Charges on oil droplets are 0.3C, 0.12C, and 0.18C, respectively

$$HCF = 0.06$$

$$\text{The number of electrons on A} = \frac{0.3}{0.06} = 5.$$

$$\text{The number of electrons on B} = \frac{0.12}{0.06} = 2.$$

$$\text{The number of electrons on C} = \frac{0.18}{0.06} = 3.$$

Hence, the correct option is (c).

12. Average atomic mass =

$$\frac{(x \times 75.77) + [(x + 2) \times 24.23]}{100}$$

$$= x + 0.48$$

Hence, the correct option is (a).

13. Deuterium nucleus has 1 proton and 1 neutron. Therefore, deuteron has 1 proton and 1 neutron. It has unit positive charge.

$$\text{The charge on deuteron} = 1.6 \times 10^{-19} \text{ C.}$$

$$\text{The mass of deuteron} = 6.7 \times 10^{-27} \text{ kg} + 7.2 \times 10^{-27} \text{ kg}$$

$$= 13.9 \times 10^{-27} \text{ kg.}$$

$$\text{The specific charge of deuteron} = \frac{1.6 \times 10^{-19}}{13.9 \times 10^{-27}}$$

$$= 1.15 \times 10^7 \text{ C/kg}$$

Hence, the correct option is (b).

14. Atomic number of element decreases by 2 units and mass number of the element decreases by 4 units.

Hence, the correct option is (b).

15. High voltage and low pressure are required to be maintained in order to increase the kinetic energy of the dislodged electrons so that the number of effective collisions (collision that can dislodge electron from the atom) increases. Finally, a proper glow can be observed on the fluorescent screen.

Hence, the correct option is (d).

CHAPTER 3

Periodic Classification of Elements

Assessment Test I

- Element X belongs to VII-A group and third period. Third period elements are called typical elements.
Hence, the correct option is (a).
- Oxidizing property is directly proportional to electro negativity and electron affinity. Therefore, fluorine is a good oxidizing agent.
Hence, the correct option is (c).
- According to the law of triads, the average of atomic weights of Be and Ca is equal to the atomic weight of Mg.
Hence, the correct option is (b).
- $A \rightarrow \text{He}$, $B \rightarrow \text{Cs}$, $C \rightarrow \text{Cl}$, $D \rightarrow \text{F}$
Maximum metallic character is Cs.
Maximum electronegativity is F.
Maximum electron affinity is Cl.
Maximum ionization energy is He.
Hence, the correct option is (a).
- $A \rightarrow \text{I-A group}$ and cation formed is A^{+1} (Na^{+1}).
 $B \rightarrow \text{II-A group}$ and cation formed is B^{+2} (Mg^{+2}).
 $C \rightarrow \text{VII-A group}$ and anion formed is C^{-1} ($\overline{\text{Cl}}$).
 $D \rightarrow \text{VI-A group}$ and anion formed is D^{-2} (S^{-2}).
 $\therefore$ Increasing order of sizes is CDBA.
Hence, the correct option is (c).
- Caesium belongs to very long period.
Xeon belongs to second long period.
Potassium belongs to long period.
Radium belongs to incomplete period.
Hence, the correct option is (b).
- Across the period, ionization energy increases due to decrease in atomic size. Due to stable valence electronic configuration of II-A group elements, they possess higher I.P values than corresponding III-A group elements.
Hence, the correct option is (b).
- Across the period, electron affinity increases due to decrease in atomic size. The 18th group elements possess zero electron affinity values due to stable octet configuration.
Hence, the correct option is (a).
- Drastic increase in the second and fourth I.P values indicates the two elements A and B belong to I-A and III-A groups, respectively.
Hence, the correct option is (a).
- Electronic configuration of element belongs to the third period and IV-A group is 2, 8, and 4.
 $\therefore$ Difference in electrons = $8 - 4 = 4$
Hence, the correct option is (c).
- $A \rightarrow 2, 8, 8, 1 \Rightarrow$ 4th period and I-A group
 $B \rightarrow 2, 8, 6 \Rightarrow$ 3rd period VI-A group
 $C \rightarrow 2, 8, 1 \Rightarrow$ 3rd period I-A group
 $D \rightarrow 2, 5 \Rightarrow$ 2nd period and V-A group
 $E \rightarrow 2, 8, 18, 8, 2 \Rightarrow$ 5th period II-A group
The correct order of atomic sizes is $E > A > C > B > D$.
Hence, the correct option is (b).
- Element A is Ba and B is Rb. I.P of Ba is less than that of Rb.
Hence, the correct option is (d).
- In case of iso electronic species, with increase in charge of anion, effective nuclear charge decreases and hence, ionic size increases. In case of cations it is exactly reverse.
Hence, the correct option is (d).
- $A \rightarrow \text{Al}$ and belongs to III-A group and 3rd period.
 $B \rightarrow \text{Ge}$ and belongs to IV-A group and 4th period.
 $C \rightarrow \text{Si}$ and belongs to IV-A group and 3rd period.

Across the period, non-metallic character increases and down the group, it decreases.

Hence, the correct option is (c).

15. X is iodine and it belongs to VII-A group.

Hence, the correct option is (d).

Assessment Test II

1. 5th period is called second long period. Elements with atomic numbers from 37 to 54 are in 5th period.

Hence, the correct option is (c).

2. Reducing property is maximum for caesium.

Hence, the correct option is (c).

3. The three triads which obey Dobereiner's Law of Triads are $(_{20}\text{Ca}, _{38}\text{Sr}, _{56}\text{Ba})$, $(_3\text{Li}, _{11}\text{Na}, _{19}\text{K})$ and $(_{17}\text{Cl}, _{35}\text{Br}, _{53}\text{I})$.

Hence, the correct option is (d).

4. $\text{A} \rightarrow \text{Na}$; Na has less tendency to form cation.

$\text{B} \rightarrow \text{Ar}$; Ar has zero electron affinity.

$\text{C} \rightarrow \text{Cl}$; Cl has good oxidizing property.

$\text{D} \rightarrow \text{Rb}$; Rb has minimum ionization energy.

Hence, the correct option is (d).

5. A belongs to IA group.

B belongs to II-A group.

C belongs to VII-A group.

D belongs to VI-A/IV-A/V-A groups.

$\therefore$ The correct order of atomic sizes is CDBA.

Hence, the correct option is (b).

6.	Electronic Configuration	Group	Type of Element
	2, 8, 8, 1	I-A	Alkali Metal
	2, 8, 18, 8, 2	II-A	Alkaline earth metal
	2, 8, 18, 18, 7	VII-A	Halogen
	2, 8, 18, 6	VI-A	Chalcogen

Hence, the correct option is (a).

7. Na^+ ion possesses a stable octet configuration.

Hence, the second I.P of Na^{+1} is greater than

that of Mg^{+1} .

Hence, the correct option is (a).

8. With increase in number of electrons in the species, the repulsive force increases. Hence, energy absorbed is more than energy released during the addition of second electron.

Hence, the correct option is (a).

9. Due to stable electronic configuration of N, it possesses high I.P value than that of C.

$\therefore \text{X} \rightarrow \text{C}$

$\text{Y} \rightarrow \text{N}$

$\text{Z} \rightarrow \text{O}$

Hence, the correct option is (b).

10. Electronic configuration of X is 2, 8, 18, and 6. The difference in the number of electrons present between the last shell and first shell is 4.

Hence, the correct option is (b).

11. Across the period, the atomic sizes decrease up to VII-A group and then increase.

Hence, the correct option is (b).

12. X, Y, and Z are Ca, Sr, and Ba, respectively. Down the group oxidizing property decreases and reducing property increases. Hence, Ba is good reducing agent.

Hence, the correct option is (a).

13. In case of cation, effective nuclear charge is more as against the corresponding neutral atom. Hence, the size of the atom is greater than cation. It is reverse in case of anions.

Hence, the correct option is (a).

14. $\text{X} \rightarrow \text{S}$, $\text{B} \rightarrow \text{O}$, $\text{D} \rightarrow \text{Se}$

$\text{A} \rightarrow \text{P}$, $\text{C} \rightarrow \text{Cl}$

Selenium is least non-metallic in nature.

Hence, the correct option is (a).

15. Phosphorus is used for manufacture of match sticks. Phosphorus belongs to the third period and V-A group.

Hence, the correct option is (a).

Assessment Test III

1. Elements are arranged in the increasing order of atomic weights in Mendeleev's periodic table.

Hence, the correct option is (a).

2. Let Z be the atomic weight of the third element.

$$\therefore \frac{x+z}{2} = y \therefore z = 2y - x.$$

Hence, the correct option is (d).

3. $30 \rightarrow 2, 8, 18, 2 \rightarrow 4\text{th period, } 12\text{th group}.$

Hence, the correct option is (b).

4. Transition metals have the highest density as they possess smaller size and larger mass.

Hence, the correct option is (c).

5. Li, being an alkali metal, is the most electropositive metal.

Hence, the correct option is (a).

6. Atomic number 19 $\rightarrow 2, 8, 8, 1$ (group 1)

Atomic number 15 $\rightarrow 2, 8, 5$ (group 15)

Atomic number 16 $\rightarrow 2, 8, 6$ (group 16)

Atomic number 9 $\rightarrow 2, 7$ (group 17)

The order is ABCD.

Hence, the correct option is (a).

7. The atomic size of an element is inversely proportional to its effective nuclear charge. This is because increase in effective nuclear charge increases the nuclear force of attraction on the outermost shell and hence, the lesser atomic size.

Hence, the correct option is (d).

8. Ionization energies of representative elements generally increase along a period since differentiating electrons enter the same shell and thereby the effective nuclear charges increase.

Hence, the correct option is (a).

9. Fluorine is the best oxidizing agent as it has a strong tendency to take up an electron to form F^- ion. This means that it undergoes reduction readily and hence, has maximum ability to oxidize the other substance.

Hence, the correct option is (a).

10. Cl, being the most electronegative among the given elements, has the highest non-metallic character.

Hence, the correct option is (a).

11. The drastic increase in the value of ionization potential implies that the stable ion is formed which cannot lose electrons further. Since z is much higher than y , we can say that the element attains stability after losing the second electron. Therefore, it has 2 valence electrons.

Hence, the correct option is (b).

12. In case of X, effective nuclear force of attraction on the valence electrons is the least and hence, it has the largest atomic radius. In case of positive ions. The nuclear charge remains the same, but the number of electrons is less. Therefore, with the increase in positive charge, the radius decreases.

Hence, the correct option is (a).

13. A good reducing agent is the one which loses electrons readily. This means that it undergoes oxidation and reduces the other substance. It loses electron(s) to attain stability. It cannot gain electrons.

Hence, the correct option is (d).

14. It is a transition metal. Group 3 to 12 are transition metals.

Hence, the correct option is (c).

15. (i) $\rightarrow (C) \rightarrow (2, 8, 7; \text{chlorine})$

(ii) $\rightarrow (D) \rightarrow (2, 8, 18, 8; \text{krypton})$

(iii) $\rightarrow (A) \rightarrow (2, 8, 8, 1; \text{potassium})$

(iv) $\rightarrow (B) \rightarrow (2, 8, 18, 1; \text{copper}).$

Hence, the correct option is (b).

Assessment Test IV

1. Atomic number is considered as the most fundamental property in the construction of the modern periodic table.

Hence, the correct option is (b).

2. $y = \frac{x+z}{2} \quad 2y = x+z$

Hence, the correct option is (b).

3. 4th period, 15th group $\rightarrow 2, 8, 18, 5$

$\therefore$ Atomic number = 33

Hence, the correct option is (c)

4. K has the least density among the given elements. Density decreases down the group and increases across a period.

Hence, the correct option is (a).

5. Electropositivity is pertinent to the tendency of an element to lose an electron.

Hence, the correct option is (b).

6. Atomic number 13 $\rightarrow 2, 8, 3$; group is 13.

Atomic number 20 $\rightarrow 2, 8, 8, 2$; group is 2.

Atomic number 6 $\rightarrow 2, 4$; group is 14.

Atomic number 14 $\rightarrow 2, 8, 4$; group is 14.

Hence, the correct option is (b).

7. Atomic number: 7 $\rightarrow 2, 5$

Atomic number: 3 $\rightarrow 2, 1$

Atomic number: 9 $\rightarrow 2, 7$

Atomic number: 11 $\rightarrow 2, 8, 1$

Decreasing order of atomic size is DBAC.

Hence, the correct option is (b).

8. Since K has the largest size among the given elements, its ionization energy is the least.

Hence, the correct option is (a).

9. Elements with greater electron affinity can take away electrons from other elements and thus the other elements undergo oxidation.

Hence, the correct option is (c).

10. Non-metallic character of an element is directly proportional to the electronegativity.

Hence, the correct option is (c).

11. Sodium has only one valence electron. Hence, it becomes stable after losing one electron. Therefore, the value of 2nd ionization potential increases drastically. Hence, the difference between the second and the first ionization energies is the maximum.

Hence, the correct option is (a).

12. They are isoelectronic with 10 electrons in their atoms. O^{-2} has the largest radius since the number of protons in it is the least. That is, the effective nuclear force of attraction on the valence electrons is less.

Hence, the correct option is (a).

13. Reducing character is the tendency to undergo oxidation and is directly proportional to the tendency to lose electrons.

Hence, the correct option is (c).

14. A non-metal occupies the right portion of the periodic table as the non-metallic character increases from left to right across a period.

Hence, the correct option is (b).

15. (i) 14 (Silicon) $\rightarrow D$

(ii) 37 (Rubidium) $\rightarrow A$

(iii) 4 (Beryllium) $\rightarrow B$

(iv) 8 (Oxygen) $\rightarrow C$

Hence, the correct option is (c).

CHAPTER 4

Chemical Bonding

Assessment Test I

1. AlCl_3 is a polar covalent compound.
Hence, the correct option is (c).
2. Since A and B belong to the 16th and 17th groups, the molecule is a polar covalent molecule. Hence, the force of attraction among them is dipole-dipole interaction.
Hence, the correct option is (a).
3. In hydrochloric acid, H_3O^+ ions are present. In H_3O^+ , dative bond is present between H_2O and H^+ .
Hence, the correct option is (d).
4. In BCl_3 , boron (B) has six electrons in its valence shell.
Hence, the correct option is (c).
5. Significant difference in ionization energy is the most suitable among the given conditions for the formation of an ionic bond.
Hence, the correct option is (a).
6. Oxidation is the loss of electrons. Therefore, A^+ to A^{++} is oxidation.
Hence, the correct option is (a).
7. Co-existence of water and water vapour in nature is not the effect of H-bond.
Hence, the correct option is (d).
8. $\text{Fe} + \text{S} \rightarrow \text{FeS}$ is a redox reaction.
Hence, the correct option is (a).
9. The bond present in CH_2Cl_2 is a polar covalent bond because all the four bonds belong to non-identical atoms.
Hence, the correct option is (a).
10. The strength of an ionic bond is directly proportional to the charge possessed by the ions and

inversely proportional to the distance between them. Hence, the order is $\text{D} < \text{A} < \text{B} < \text{C}$.

Hence, the correct option is (c).

11.	Molecule	No. of Covalent Bonds
	N_2	3
	O_2	2
	H_2	1
	CO_2	4

$\therefore$ The order is CBAD.

Hence, the correct option is (d).

12. Metallic bond in calcium is stronger than in potassium because the charge possessed by Ca^{+2} ion is more than K^+ and the radius of Ca^{+2} is smaller than K^+ .
Hence, the correct option is (a).
13. Since metals are electropositive in nature, they have the tendency of losing electrons. Therefore, they are good reducing agents.
Hence, the correct option is (d).
14. (i) A has 3 protons and B has 17 protons.
 $\text{AB} \rightarrow (\text{C})$
(ii) A has 8 protons and B has 9 protons.
 $\text{AB}_2 \rightarrow (\text{D})$
(iii) A has 17 protons and B has 8 protons.
 $\text{A}_2\text{B} \rightarrow (\text{A})$
(iv) A has 5 protons and B has 9 protons.
 $\text{AB}_3 \rightarrow (\text{B})$
Hence, the correct option is (b).
15. Ionic compounds conduct electricity in molten state.
Hence, the correct option is (a).

Assessment Test II

1. KCl is an electrovalent compound.
Hence, the correct option is (d).
2. H-bonding exists in AH_3 where A is nitrogen according to the description given.
Hence, the correct option is (d).
3. In NH_4Cl , dative bond is present between NH_3 and H^+ .
Hence, the correct option is (b).
4. In SnCl_4 , Sn has 8 electrons in its valence shell, whereas the other given atoms have 10 electrons.
Hence, the correct option is (b).
5. Both the atoms participating in the formation of covalent bond should be non-metals.
Hence, the correct option is (a).
6. A to A^+ is oxidation. All other options signify reduction.
Hence, the correct option is (a).
7. Existence of water in the liquid state is the effect of H-bond.
Hence, the correct option is (a).
8. $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$ is not a redox reaction.
Hence, the correct option is (a).
9. In HClO , polar covalent bond is present between H and Cl and dative bond is present between Cl and O.
Hence, the correct option is (d).
10. The strength of ionic bond is directly proportional to the charge possessed by the ions and inversely proportional to the distance between them. The order is $\text{SrO} > \text{BaO} > \text{Cs}_2\text{O} > \text{CsCl}$.
Hence, the correct option is (a).

11.	Molecule	No. of Covalent Bonds
	C_2H_2	5
	CHCl_3	4
	NH_3	3
	Cl_2	1

The order is ABCD.

Hence, the correct option is (c).

12. Sodium can be cut with a knife because the size of Na^+ is bigger and it is monopositive. Hence, the strength of the metallic bond is less.
Hence, the correct option is (c).
13. Non-metals are good oxidizing agents because electron affinity increases with the increase of the non-metallic character.
Hence, the correct option is (c).
14. (i) A has 7 protons and B has 1 proton. $\rightarrow$ (C) NH_3 (H-bonding)
(ii) Both the atoms have identical electronic configurations. $\rightarrow$ (D) Non-polar covalent molecule (Van der Waals' forces of attraction)
(iii) A has 20 protons and B has 8 protons. $\rightarrow$ (A) Interionic attraction
(iv) A has 1 proton and B has 16 protons. $\rightarrow$ (B) $\text{H}_2\text{S} \rightarrow$ Dipole-dipole interaction
Hence, the correct option is (a).
15. Polar covalent compounds ionize in water, but non-polar covalent compounds do not.
Hence, the correct option is (a).

Assessment Test III

1. IF_7 is a stable halide of iodine. However, iodine does not attain octet configuration in this molecule. It has 14 electrons around it.
Hence, the correct option is (b).
2. Nitrogen and chlorine have equal electronegativity values. However, ammonia has hydrogen bonding due to the small size of nitrogen atom when compared to the size of chlorine atom.
Hence, the correct option is (c).
3. Ease of formation of ionic chlorides depends upon the ionization potential of the metal. The lower the IP value of a metal, the easier is the cation formation. The IP value decreases from top to bottom in a group and increases from

left to right in a period. Therefore, the required order is ECDAB.

Hence, the correct option is (a).

4. The strength of an ionic bond is directly proportional to the charges of the ions and inversely proportional to their sizes.

Therefore, the required order is CBAD.

Hence, the correct option is (d).

5. (i) $\rightarrow$ (C)

(ii) $\rightarrow$ (D)

(iii) $\rightarrow$ (E)

(iv) $\rightarrow$ (A)

(v) $\rightarrow$ (B)

Hence, the correct option is (d).

6. $\text{H}_2\text{S} + \text{SO}_2 \rightarrow \text{H}_2\text{O} + \text{S}$ is a redox reaction, since it involves a change in the oxidation state of sulphur.

Hence, the correct option is (c).

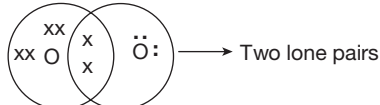
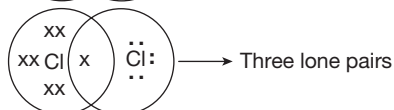
7. Electrovalence of metal is 3 as it has 3 electrons in its valence shell and it loses them to form a trivalent cation. Electrovalence of the non-metal is 1 as it has 7 electrons in its valence shell and gains one electron to form a uninegative ion.

Hence, the correct option is (b).

8. The electronic configurations of calcium and bromine are 2, 8, 8, 2 and 2, 8, 18, 7, respectively. Calcium loses 2 electrons to get argon configuration; bromine gains one electron to get krypton configuration.

Hence, the correct option is (d).

9. $\text{H}:\text{H} \rightarrow$ no lone pairs



Hence, the correct option is (b).

10. $Z = 17$; electronic configuration 2, 8, 7

$Z = 8$; electronic configuration 2, 6

Since both are non-metals, they are involved in the formation of a covalent compound.

Hence, the correct option is (b).

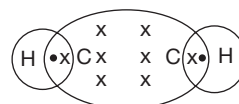
11. Malleability, metallic lustre, and thermal conductivity are the characteristic properties of metals and they occur due to the presence of metallic bonds. The reducing property of metals is due to their low IP values.

Hence, the correct option is (c).

12. Ionic bond, covalent bond, and dative bonds are bonds formed between atoms within the molecules. Hydrogen bonding is a stronger, dipole-dipole attraction present between the molecules due to high polarity.

Hence, the correct option is (d).

13. Acetylene molecule can be represented by the following Lewis dot formula.



Thus, there is a triple covalent bond between the carbon atoms.

Hence, the correct option is (c).

14. The element with electronic configuration 2, 3 is boron. The element with electronic configuration 2, 7 is fluorine. The compound formed is BF_3 . In BF_3 , the central atom possesses contracted octet. It is an electron-deficient molecule and can act as an electron acceptor. After bonding fluorine, it has 3 lone pairs around it. Although boron does not attain octet in this molecule, it is a stable molecule.

Hence, the correct option is (d).

15. NH_3 has covalent bonds between nitrogen and hydrogen. There is a dative bond between NH_3 and H^+ . Ionic bond exists between NH_4^+ and OH^- . The same can be applied to ammonium chloride. Blue vitriol is $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$. Ionic bond is present between Cu^{+2} and SO_4^{2-} . Water molecules are bonded to Cu^{+2} ions by dative bonds. Therefore, all the molecules possess all

the three types of bonds.

Hence, the correct option is (d).

Assessment Test IV

1. Carbon attains neon configuration by sharing four electrons with other atoms. Hence, it exhibits tetravalency.

Hence, the correct option is (a).

2. H_2O has a higher boiling point than HF due to greater number of hydrogen bonds formed in case of water, even though HF has a stronger hydrogen bond.

Hence, the correct option is (d).

3. $Z = 8$; oxygen
 $Z = 17$; chlorine
 $Z = 9$; fluorine
 $Z = 7$; nitrogen

The ease of formation of anion depends upon the electron affinity value of non-metals. Electron affinity increases from left to right in a period and decreases from top to bottom in a group. Thus, the correct order is $\text{Cl} > \text{F} > \text{O} > \text{N}$. Therefore, the required order is CDAB.

Hence, the correct option is (c).

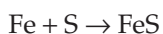
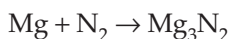
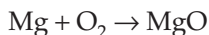
4. The strength of the ionic bond is inversely proportional to the size of the anion, provided the cation is the same. The correct decreasing order of strength ionic compounds should be DBEAC.

Hence, the correct option is (a).

5. (i) $\rightarrow$ (C)
 (ii) $\rightarrow$ (D)
 (iii) $\rightarrow$ (A)
 (iv) $\rightarrow$ (B)
 (v) $\rightarrow$ (E)

Hence, the correct option is (d).

6. $\text{M} + \text{acid} \rightarrow \text{metal salt} + \text{H}_2$



All these reactions are redox reactions.

Hence, the correct option is (d).

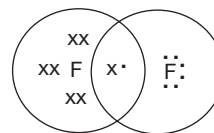
7. In M_2X_3 , metal has +3 charge and non-metal has -2 charge. Therefore, the electrovalencies of metal and non-metal are 3 and 2, respectively.

Hence, the correct option is (c).

8. The electronic configuration of metal could be 2, 8, 1. It can lose an electron to get the configuration of neon. The electronic configuration of a non-metal could be 2, 5 as it can gain three electrons to get the configuration of neon.

Hence, the correct option is (d).

9. The element with electronic configuration 2, 7 is fluorine. The Lewis dot diagram for F_2 molecule is as follows.



There are 3 lone pairs on each fluorine atom.

Hence, the correct option is (c).

10. $Z = 16$ corresponds to sulphur and $Z = 8$ corresponds to oxygen. Sulphur and oxygen belong to VI group and form a covalent compound.

Hence, the correct option is (a).

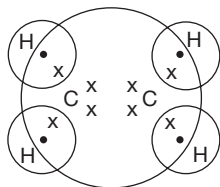
11. In iron, the penultimate shell electrons can also participate in metallic bonding. As more number of electrons are involved in metallic bonding, the metallic bond becomes strong. This imparts greater hardness to iron.

Hence, the correct option is (d).

12. Hydrogen bonding is responsible for relatively high boiling point of water and dissolution of glucose in water. Due to the presence of hydrogen bonding, water exists as associated liquids. Conduction of electricity does not depend on hydrogen bonding. It depends on the availability of free electrons or ions as charge carriers.

Hence, the correct option is (d).

13. Ethylene molecule can be represented by the following Lewis dot diagram.



It shows the presence of 6 bond pairs in the molecule.

Hence, the correct option is (d).

14. The element with electronic configuration 2, 8, 6 is sulphur. The element with electronic

configuration 2, 7 is fluorine. The compound formed is SF_6 . Sulphur has 6 electrons in its valence shell and shares six electrons from 6 fluorine atoms. Therefore, in SF_6 , the central atom, sulphur acquires 12 electrons.

Hence, the correct option is (c).

15. Hydronium ion possesses dative bond between H_2O molecule and H^+ ion. Within H_2O molecule, there are covalent bonds. It does not contain ionic bond.

Hence, the correct option is (a).